

SmartSpend — AI-Powered Expense Tracker

A full-stack expense management web application built from scratch with **Java Servlets**, **MySQL**, and **vanilla HTML/CSS/JS** — featuring real-time analytics, budget tracking, and AI-powered financial advice.

Features

Feature	Description
 User Auth	Secure registration & login with session-based authentication
 Expense Management	Add, view, and delete expenses with category tagging
 Budget Tracking	Set monthly budgets with real-time progress indicators
 Dashboard Analytics	Live stats — total spent, remaining budget, top categories
 Interactive Charts	Doughnut, Bar & Line charts powered by Chart.js

Tech Stack

Backend

- Java Servlets (Jakarta EE)
- Apache Tomcat — Application Server
- MySQL — Relational Database
- JDBC — Database Connectivity

Frontend

- HTML5 / CSS3 / Vanilla JavaScript
- Chart.js — Data Visualizations

AI & APIs

- Groq API — Llama 3.3 70B (Financial Advice Engine)

Tools & IDE

- Eclipse IDE
- MySQL Workbench

PROJECT STRUCTURE

```
• SmartSpend/
•   └─ src/main/
•       └─ java/com/smartspend/
•           └─ servlet/
•               ├── LoginServlet.java
•               ├── RegisterServlet.java
•               ├── ExpenseServlet.java
•               ├── BudgetServlet.java
•               ├── DashboardServlet.java
•               └── AIAdviceServlet.java
•           └─ dao/
•               ├── UserDAO.java
•               └── ExpenseDAO.java
•           └─ model/
•               ├── User.java
•               └── Expense.java
•           └─ util/
•               └── DBConnection.java
•       └─ webapp/
•           ├── WEB-INF/web.xml
•           ├── index.html
•           ├── dashboard.html
•           ├── css/style.css
•           └─ js/
•               ├── charts.js
•               └── app.js
```

Setup & Installation

Prerequisites

- Java JDK 11 or higher
- Apache Tomcat 10+
- MySQL 8.0+
- Eclipse IDE (or any Java IDE)
- Groq API Key — console.groq.com

1.Set Up the Database

CREATE DATABASE smartspend_db;

USE smartspend_db;

```
CREATE TABLE users (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    username VARCHAR(50) NOT NULL UNIQUE,  
    email VARCHAR(100) NOT NULL UNIQUE,  
    password VARCHAR(255) NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE expenses (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    user_id INT NOT NULL,  
    title VARCHAR(100) NOT NULL,  
    amount DECIMAL(10, 2) NOT NULL,  
    category VARCHAR(50),  
    expense_date DATE NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (user_id) REFERENCES users(id)  
);
```

```
CREATE TABLE budgets (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    user_id INT NOT NULL,  
    month VARCHAR(7) NOT NULL,  
    amount DECIMAL(10, 2) NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES users(id)  
);
```

2. Configure Database Connection

```
private static final String URL = "jdbc:mysql://localhost:3306/smartspend_db";
```

```
private static final String USER = "your_mysql_username";
```

```
private static final String PASSWORD = "your_mysql_password";
```

3. Configure Groq API Key

```
private static final String GROQ_API_KEY = "your_groq_api_key_here";
```

4. Deploy on Tomcat

1 Import the project into Eclipse as a Dynamic Web Project

2 Add the project to your Tomcat server

3 Start the server

4 Visit <http://localhost:8080/smartspend>